



Hall Ticket No-

Code: S123528

Stanley College of Engineering and Technology for Women(A)**B.E(CSE/EEE/ECE/IT/CME/AI&DS) II Semester (Backlog) Examinations May-June-2023****Mathematics-II**

Time: 3 hours

Max.Marks:60

PART-A**Note: Answer all questions (Compulsory)****5 X 2=10M**

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|---|--|-----|----|
| 1 | Write standard form of Bernoulli's differential equation of first order. | 2 M | L1 |
| 2 | Find the Particular Integral of $(D^2 + 4^2)y = e^{-4x}$ | 2 M | L1 |
| 3 | Define the Gamma function. | 2 M | L1 |
| 4 | State Cayley-Hamilton theorem. | 2 M | L1 |
| 5 | Find the Laplace Transform of t^3 | 2 M | L1 |

PART-B**Note: Answer all questions****5 X 10= 50M**

- | | | | | |
|-----------|----|--|-----|----|
| 6 | a. | Find the Orthogonal Trajectory of the family of curves $r^n = a^n \cos n\theta$, where θ is parameter. | 5M | L4 |
| | b. | Solve $(3x^2 + 6xy^2)dx + (6x^2y + 4y^3)dy = 0$ | 5M | L3 |
| OR | | | | |
| | c. | Solve $(3y^2 + 4x) dx + 3xydy=0$ | 5M | L4 |
| | d. | Solve $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$ | 5M | L3 |
| 7 | a. | Solve $\frac{d^2x}{dt^2} + 5 \frac{dx}{dt} + 6x = 0$, given $x(0) = 0, \frac{dx}{dt}(0) = 15$ | 5M | L3 |
| | b. | Solve $(D^2 - 6D + 25)y = e^{2x}$. | 5M | L3 |
| OR | | | | |
| | c. | Solve by the method of variation of parameters, $\frac{d^2y}{dx^2} + 4y = \tan 2x$. | 10M | L3 |
| 8 | a. | Show that $\int_a^b (x-a)^m (b-x)^n dx = (b-a)^{m+n-1} \cdot \beta(m+1, n+1)$ | 5M | L3 |
| | b. | Show that $\int_0^\infty \sqrt{x} e^{-x^3} dx = \frac{\sqrt{\pi}}{3}$ | 5M | L3 |
| OR | | | | |
| | c. | State and Prove the relation between Beta and Gamma. | 5M | L3 |
| | d. | Prove that $\beta\left(m, \frac{1}{2}\right) = 2^{2m-1} \beta(m, m)$ | 5M | L3 |

- 9 a. Reduce the quadratic form into canonical form by an orthogonal transformation. Also state the nature of quadratic form

10M L2

$$8x_1^2 + 7x_2^2 + 3x_3^2 - 12x_1x_2 - x_2x_3 + 4x_3x_1$$

OR

- c. Find the characteristic equation of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence compute A^{-1} .

5M L2

- d. Find the Eigen values of $\text{adj } A$ and $A^2 - 2A + I$, where $A = \begin{pmatrix} 2 & 3 & 4 \\ 0 & 4 & 2 \\ 0 & 0 & 3 \end{pmatrix}$

5M L2

- 10 a. Find the Laplace transform of $t^3 e^{-3t}$.

5M L3

- b. Find the inverse Laplace transform of the $\frac{1}{s(s^2+a^2)}$ by convolution theorem.

5M L3

OR

- c. Solve, using Laplace transform method, $y''' + 2y'' - y' - 2y = 0$ with the conditions $y(0) = y'(0) = 0$ & $y''(0) = 6$

10M L3



Hall Ticket No:

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Stanley College of Engineering and Technology for Women (A)

B.E (ECE) II Semester (Backlog) Examinations May-June-2023

Circuit Theory

Time: 03Hours

Max.Marks:60

Part- A

Note: Answer all questions (Compulsory)

5X2=10M

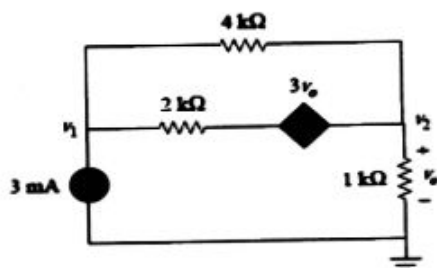
1. Distinguish between super node and super mesh. 2M L1
2. List any two differences between transient response and steady state response. 2M L1
3. Describe the purpose of Brune test in two port networks? 2M L1
4. Define Q factor. 2M L1
5. Identify the difference between constant k and m derived filters. 2M L1

Part- B

Note: Answer all questions

- 6.a Calculate v_1 and v_2 in circuit shown, using nodal analysis.

5M L3



- b. Demonstrate the principle of Duality with an example.

5M L3

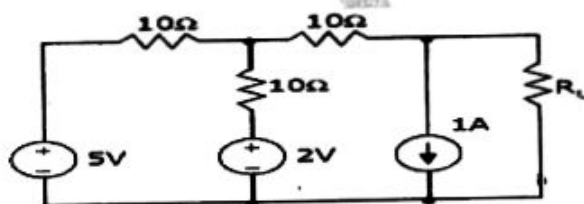
OR

- c. Demonstrate superposition theorem with suitable example.

5M L3

- d. Find the value of R_L value such that maximum power is delivered to it and also calculate maximum power delivered to it.

5M L3



7. a. Derive the response of a series RLC circuit for a step input.

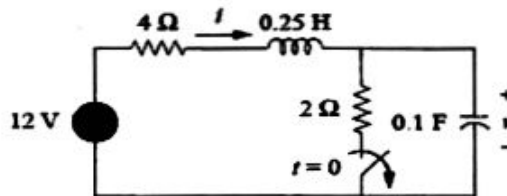
5M L3

- b. What is real power? What is reactive power? Which power, real or reactive, has an average value of zero?

5M L2

OR

- c. Explain concept of Phasors with suitable example. 5M L2
- d. The switch in following figure has been closed for long and is open at $t=0$. Find 5M L3
 i) $i(0+)$ ii) $i(\infty)$ iii) $v(0+)$ iv) $v(\infty)$



8. a. Obtain the relation between Y and Z parameters. 5M L4
- b. Explain the symmetry and reciprocity conditions for ABCD parameters and h-parameters. 5M L2
- OR**
- c. Explain driving point function and transfer function in two port networks. 5M L2
- d. Two two-port networks are connected in parallel. Prove that the overall y parameters are the sum of corresponding individual y-parameters. 5M L4
9. a. Illustrate with an example how to find natural response from pole zero plot. 5M L4
- b. A capacitor of reactance 5Ω is connected in series with a 10Ω resistor. The whole circuit is then connected in parallel with a coil of inductive reactance 20 and a variable resistor. Measure the value of this resistance for which the parallel network is resonant. 5M L5
- OR**
- c. Outline a procedure to find an expression for resonant frequency and bandwidth in series RLC circuit. 5M L4
- d. A series R-L-C circuit comprises a $5\mu\text{F}$ capacitor, a 4Ω resistor and a variable inductance L. The supply voltage is $10\angle 0^\circ$ V at a frequency of 159.1 Hz. The inductance is adjusted until the voltage across the 4Ω resistance is a maximum. Select the value of inductance and measure the Q-factor of the circuit. 5M L5
10. a. Explain the design of constant K band pass filters. 5M L2
- b. Design a m-derived low filter with a design impedance of 300Ω and the cut off frequency at 2 KHz and infinite attenuation at 2345Hz. 5M L6
- OR**
- c. Design a constant K band pass filter with cut-off frequencies of 2000Hz and 5000Hz with a design impedance of 500 ohms. 5M L6
- d. Explain pass band, stop band and attenuation in filters. 5M L2

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Stanley College of Engineering and Technology for Women(A)

B.E(ECE & EEE) II Semester (Backlog) Examinations May-June-2023

Engineering Physics

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

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|---|---------|
| 1 Classify Fraunhofer and Fresnel diffraction patterns. | 2 M BL2 |
| 2 List any four applications of Lasers. | 2 M BL1 |
| 3 Define Unit cell and Bravais lattice. | 2 M BL1 |
| 4 What is de-broglie's concept of matter waves? | 2 M BL1 |
| 5 What is Biot-savart's law? | 2 M BL1 |

PART-B

Note: Answer all questions

5 X 10= 50M

- | | |
|--|--------|
| 6 a. Explain the formation of Newton's rings. | 4M BL4 |
| b. Discuss the conditions for maxima and minima for diffraction at single slit. | 6M BL6 |
| OR | |
| c. Classify the interference of light by division of amplitude and by division of wavefront. | 4M BL2 |
| d. Explain the distribution of intensity due to N slit grating. | 6M BL2 |
| 7 a. Explain the construction and working of He-Ne laser. | 6M BL2 |
| b. Explain the concept of numerical aperture in fiber optics. | 4M BL2 |
| OR | |
| c. Explain the characteristics of lasers. | 4M BL2 |
| d. Explain Double Crucible method to produce optical fiber. | 6M BL2 |
| 8 a. Explain Bravais lattices in crystal systems. | 6M BL2 |
| b. Explain line defects and plane defects that occur in crystals. | 4M BL2 |
| OR | |
| c. Develop an expression for the concentration of Schottky defects in ionic crystals. | 6M BL3 |
| d. How Miller indices are used to identify the orientation of planes in crystals? | 4M BL1 |
| 9 a. Develop Schrodinger time independent wave equation. | 6M BL3 |
| b. Explain electric polarization in dielectrics. | 4M BL1 |
| OR | |
| c. Develop an expression for wavefunction associated with a particle in 1-D box. | 6M BL3 |
| d. Explain Gauss law of electricity. | 4M BL2 |
| 10 a. Build an equation for plane electromagnetic wave in free space. | 6M BL3 |
| b. Explain magnetic flux density and give Maxwell's equation | 4M BL2 |
| OR | |
| c. Develop Maxwell's equation in integral and differential form for Ampere's law. | 6M BL3 |
| d. Explain Faraday's law of electromagnetic induction. | 4M BL2 |
